Biasing:

Start with bias trimmer R8 set to minimum resistance.
Power up with no load first: transformer disconnected,
input shorted to ground, and current limiting if available.

Measure DC millivolts across R1 and R2.
Because R1 and R2 are 1Ω, 1mV = 1mA.

Aim for 50mV–80mV across each resistor.
Let the amp warm up for 10–15 minutes.
Measure again and adjust if needed.

Negative feedback (NFB pot):

Counter-clockwise = LOOSE,
highest feedback resistance,
(slightly above 1M, R18 + R19 + NFB pot)
least negative feedback,
lowest damping.

Clockwise = TIGHT,
lowest feedback resistance
(94k, the sum of R18 and R19),
maximum negative feedback,
highest damping.

Total input impedance is around 40k.
You need a preamp/buffer in front of the amp.

Put R4 and R5 as close to the gate pins as possible.
Also put the zener diodes D2, D3, D4, and D5 as close
to the MOSFET pins as possible.

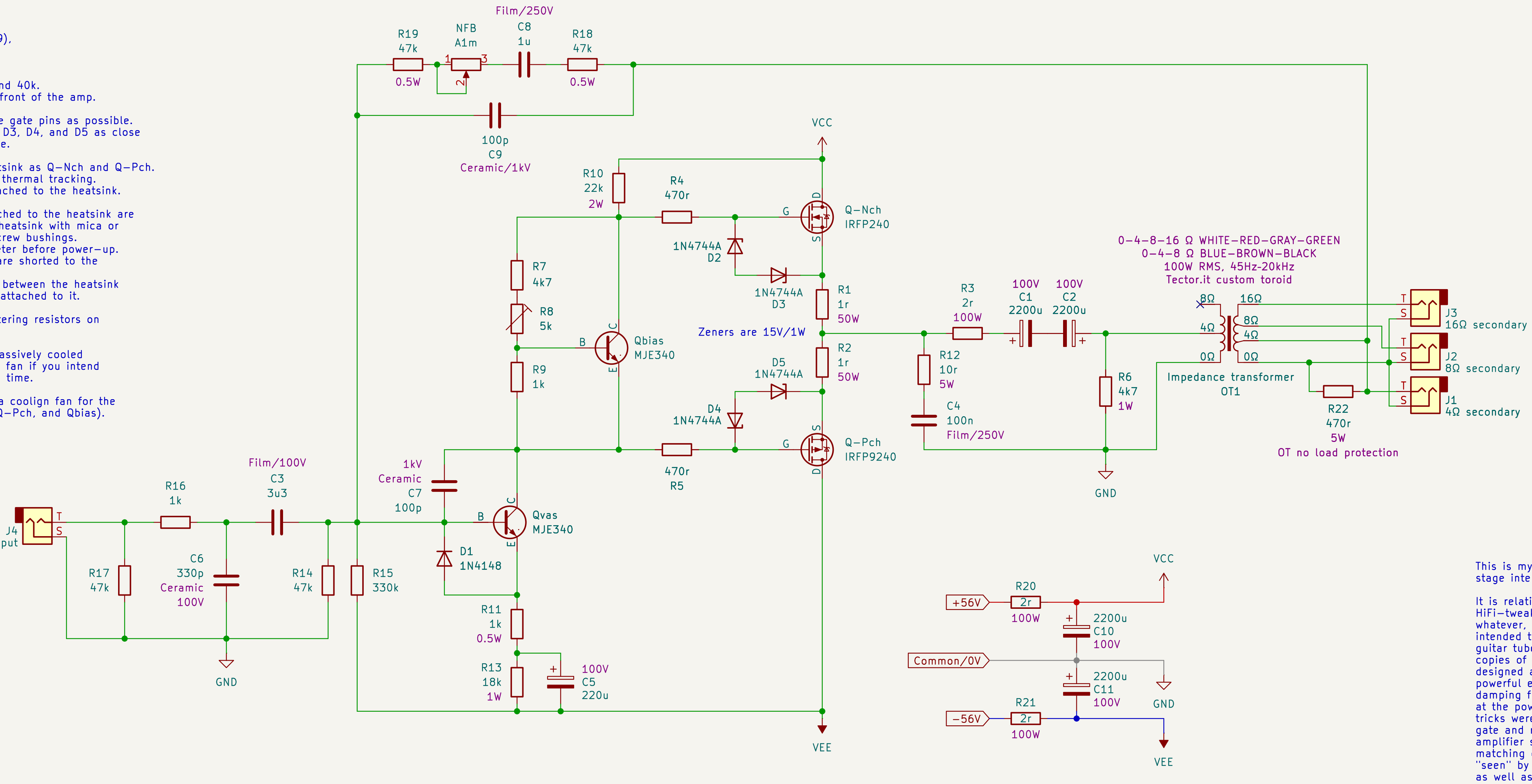
Attach Qbias to the same heatsink as Q–Nch and Q–Pch.
Put it between them; it needs thermal tracking.
Qvas does not need to be attached to the heatsink.

Make sure all transistors attached to the heatsink are
electrically isolated from the heatsink with mica or
silicone pads and insulated screw bushings.
Verify isolation with a multimeter before power–up.
Make sure no transistor legs are shorted to the
heatsink.
There should be no continuity between the heatsink
and any leg of any transistor attached to it.

Install R20 and R21 power filtering resistors on
a passively cooled heatsink.

Install R1, R2, and R3 on a passively cooled
heatsink. Maybe add a cooling fan if you intend
to push the amps hard all the time.

I strongly recommend to add a coolign fan for the
transistors heatsink (Q–Nch, Q–Pch, and Qbias).



This is my own discrete Class AB amplifier design with MOSFET push–pull output stage intended for guitar amplification.

It is relatively simple and straightforward. I intentionally avoided any HiFi–tweaks (like differential input, "current source", "current mirror" or whatever, I only used Vbe multiplier here to bias the MOSFETs), the amp is intended to be imperfect, having damping factor less than 1 (similar to a guitar tube amp). This is a single amplifier design with an intention to use 2 copies of it in a bridge–mode configuration. So damping factor target value is designed around the idea it's going to be a bridge–mode setup. The amp must be powerful enough (100W in bridge–mode, after all the losses on the output damping factor reduction ballast resistors). It only needs to be stable enough at the power expected from it. And it needs to be reliable, so few protection tricks were applied (like zener back–to–back diodes around MOSFETs source and gate and reverse turn on/off spikes 1n4148 protection diode for the voltage amplifier stage transistor). The design relies on the presence of an impedance matching output transformer for the sake of keeping the same load impedance "seen" by the amplifier in order to keep the damping factor value predictable as well as transformer feel and coloration. It also has a global negative feedback taken from the output transformer secondary (just like in tube amplifiers), so that negative feedback has the coloration of the transformer included. With the NFB pot you can set how much LOOSE or TIGHT the amplifier has to be, which includes speaker impedance curve modulation impact, tuning the response to your taste.

The power supply has an aggressive RC filter with 2 ohms series resistor. It can make add some level of "sag"/compression for the attack transients, similar to some guitar tube amplifiers. As well as it provides nice DC filtering for the amplifier since it has poor power supply noise rejection.

For the bridge–mode configuration I intended 1:2 line balancing (to send positive to one amp and negative to the other) transformer to be used for the input signal, which also provides some gain compensation (in 1:2 configuration, since there are losses on the output damping factor reduction ballast resistors, and the amp itself does not have lots of gain). So your signal gotta be hot enough to push it to its full power. Note that the input transformer better have plenty of clean headroom to be able to handle bass content in a hot signal. For that reason I would recommend for example OEP A262A2E over TY–250P which I often use in my other builds.

WORK IN PROGRESS...
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Title: Wenzel's Powered Guitar Cabinet 2606

Size: A3 | Date: June 2026 | **Rev: r1–wip–1**
KiCad E.D.A. 9.0.6 | Id: 1/1